1. Define computer performance. Explain the factors affecting CPU performance.
2. Define Bus. Compare single-bus and multiple-bus structures.
3. What is a bus? Explain different types of buses in a computer system.
4. Explain the functions of PC, MAR, MDR, IR, and Accumulator.
5. What is the role of Program Counter and Instruction Register during instruction execution?
6. Explain the functional units of a computer with a neat block diagram
7. Define Computer Architecture and Computer Organization. Explain their differences with suitable examples.
8. What are the major factors that influence CPU performance? Explain using the basic performance equation.
9. Write the formula for MIPS and CPI. Explain why MIPS and CPI are not reliable metrics for comparing the performance of different computer systems.
10. Why are MIPS and CPI not reliable metrics for comparing the performance of different computer systems?
11. Why is geometric mean preferred over arithmetic mean when computing overall SPEC performance?
12. Why does a benchmark suite consist of multiple programs instead of a single program?
13. Differentiate between RISC and CISC.
14. Differentiate between Von-Neumann Architecture and Harvard Architecture.
15. How do pipelining and parallel processing improve performance?
16. Write the **performance equation** and explain each component.
17. How does bus width affect system performance?
18. Compare a **single bus** and **multiple bus system**.
19. Explain the differences between **data, address, and control buses**.
20. What is a **bus structure** in computer organization?
21. How do system registers affect performance?
22. Explain how data flows in the **fetch-decode-execute sequence**.

23. Define **basic operational concepts** of a computer.
24. Why registers are considered functional units?
25. Differentiate between PC and IR.
26. Differentiate between computer organization and computer architecture
27. Briefly discuss all the blocks of the functional block diagram of a computer with a neat sketch. Write down the major differences among Single-bus and Multi-bus Organization used to connect the functional units.
28. Write down the steps for instruction fetch and define what an instruction format is?
29. What role do registers play in interfacing with memory? Explain their functions during memory related operations.
30. Explain the Processor memory interconnection with diagram. Write the steps needed to execute the machine instruction ADD R1, R2, R3 (Assume R1 is the destination register)
31. Write the basic performance equation of a computer. Explain the different parameters for improving the performance.
32. Explain the processor memory interconnection diagram in detail. List the steps needed to fetch and execute the machine instruction ADD R1, LOCA, where R1 is the destination register.
33. What are the phases of instruction execution? Explain the operational concepts of instruction execution for the following instructions using the register-register operation. (Assume L.H.S. operand as destination operand).
    - ADD LOCA, LOCB
    - MUL R0, LOCA
34. Mention the functions of different processor registers
    - IR
    - MAR
    - PC
35. Write the function of the different registers MAR, MDR, PC, and GPR.
36. Why Zero address instruction format take less execution time than Three address format? Explain with example.

37. Convert the following instruction in 3-address, 1 address and 0-address instruction format along with RTN.

    M = (X-Y) *(P+Q) / (R+S)

1. A processor uses PC-relative addressing for branch instructions.

   Given: PC = 4000, Instruction length = 4 bytes, Offset field = −20 (signed value)

   Calculate:

   i. Calculate next PC after fetch.

   ii. Calculate effective branch target address.

   iii. Explain why PC-relative addressing is preferred in modern systems.

38. A processor supports the following addressing modes: Immediate, Direct, Register Indirect, Indexed, PC-Relative. Given: PC = 1000, R1 = 200, R2 = 500. Memory contents:

    M[200] = 800, M[500] = 1200, M[800] = 50, M[1200] = 75, M[1020] = 90

    For each instruction below, determine: i. Effective Address (EA) ii. Operand value

    Instructions:

    a. LOAD #40
    b) LOAD 200
    c) LOAD (R1)
    d) LOAD 20(R2)
    e) LOAD 20(PC)

39. Calculate addressable memory if address bus is 24 bits.

40. Compare RISC and CISC based on instruction format, addressing modes, and execution time.

41. What is indexed addressing mode? How is it different from relative addressing mode?

42. Define effective address.

43. Define Instruction Set Architecture (ISA).

44. What is an instruction format?

45. Define Big-endian and Little-endian representation with example.

46. A 32-bit hexadecimal number 0xCAFEBABE is stored starting at memory address 3000 in a little-endian system. Write the byte stored at addresses 3000 to 3003.

47. Represent the 32-bit hexadecimal number 0xA1B2C3D4 in big-endian format. Show the contents of memory starting from address 2000.

48. What will be content of each register after executing each of the Instruction Where R2=100.

    Move R1,(R2)

    Move R3,25(R2)

    Move R4,R2

    Move R5,-(R2)

49. What do mean by addressing mode? Why it is used? Registers R1 and R2 of a computer contains the decimal values 1200 and 4600. What is the effective address of the memory operand in each of the following instruction. a) Load R5, 20(R1) b) Move R5, #3000 C) Store 30(R1,R2),R5 c) Add R5, -(R2) d) Sub R5, (R1)+

50. An instruction is stored at location 300 with its address field at location 301. The address field has the value 400. A processor register R1 contains the number 200.Evaluate the effective address if the addressing mode of the instruction is (i) Direct addressing mode (ii) Relative addressing mode (iii)Register indirect addressing mode (iv) Index addressing mode with R1 as the index register

51. Write assembly programs and show RTN that can evaluate the expression X= (A+B) (C+D)in a machine that supports (i) Two-address instruction format (ii) one-address instruction format and (iii) Zero-address instruction format.

52. Write assembly programs and show RTN that can evaluate the expression X=(A*B+C*D)/(E-F) in a machine that supportsi() Three- address instruction format(ii) Two-address instruction format (iii) one-address instruction format and (iv) Zero-address instruction format.

53. What do mean by effective Address(EA)?

54. Define **Instruction Set Architecture (ISA).**What factors influence the design of an ISA?

55. Compare **big-endian** and **little-endian** byte order.
56. What do you mean by Addressing Mode?Briefly Explain all Addressing Modes.
57. What will be the content of each register after the execution of following instructions by considering the value of R2=100?
    MOV R1, [R2]
    MOV R3, 25[R2]
    MOV R4, R2
    MOV R5, -[R2]
    Consider the following memory contents for the calculation purpose.

Address	Content
:	
99	10
100	20
120	30
125	40
:	

58. Write the assembly code and mention the Register transfer notation (RTN) to evaluate the following arithmetic expression: Z = (A - B + C) * (D / E * F)
    i) Using an accumulator type computer with one address instructions.
    ii) Using a stack organized computer with zero-address operation instructions.
    iii) Using RISC computer instruction format.

i) Using Accumulator Type Computer (One Address Instructions)